Project Planning & Management Document

1. Project Overview

Project Title: Ferr-y Dust
Team Members & Roles: [Your Name] & [Partner's Name]
Brief Description: A real-time bio-feedback installation that translates a user's heart rate into synchronized light pulses and a mathematical pentatonic audio scale.

2. Team Roles & Responsibilities

We did not have a rigid split. We both contributed to the MicroPython debugging and the electronics assembly.

Member Name	Role	Responsibilities
Ananya Naidu		Product sketching and ideation, Circuit Assembly: Wiring the ESP32 to the pulse sensor, NeoPixel ring, and speaker. Connecting the PAM8403 amplifier to make the speaker loud enough. Setting up the Buck Converter. Troubleshooting and debugging. Coding for all components individually (Pulse sensor, Neopixel, Speaker, Motor) and testing components when combined. Physical assembly of project. Material sourcing. Material testing (Chladi plate, Ferrofluid, Magnetite powder.) Soldering final circuit.
Twisha Sawant		3D modelling in blender, Videography, Circuit Assembly: Wiring the ESP32 to the pulse sensor, NeoPixel ring, and speaker. Setting up the Buck Converter. Troubleshooting and debugging. Coding for all components individually (Pulse sensor, Neopixel, Speaker, Motor)

		and testing components when combined. Physical assembly of project. Material testing (Chladi plate, Ferrofluid, Magnetite powder.) Soldering final circuit.

3. Project Timeline & Milestones

Week	Goals	Deliverables	Status
Week 1	Research material-based bio-feedback and explore initial visualization concepts.	Finalize core concept. Map flow of the project. List materials and components we need and source everything.	Finalized, but core concept was changed at the end of the week. Materials and components ordered.
Week 2	Component sourcing and initial coding. (just testing if the individual components work)	Make each component work separately. Understand logic behind the components. Start ideating the physical setup, look for aesthetics, inspiration.	Some components did not arrive. Others were tested and they work. Outer setup is in progress.
Week 3	System integration and Troubleshooting. Finalizing the enclosure design in Blender.	Be ready with final integrated prototype, stable MicroPython script, and 3D printed enclosure.	Final assembly made. Some components were excluded due to them not working. Alternatives were tested and documented.

4. Weekly Progress Log

WEEK 1-
Planned Tasks: Initial research into bio-feedback mechanics and sourcing materials for magnetic fluid (ferrofluid) visualization.

Completed Tasks: Ideation on how to map a biological pulse to physical matter and preliminary sketches of a ferrofluid-based interface.

Challenges Faced: We quickly realized that the initial brief did not require a deep conceptual justification, and we had not prepared a solid "reason" for the project. Furthermore, sourcing ferrofluid was very difficult locally.

Solutions: We decided to move forward with the same idea, but we are going to try and see if we can make the ferrofluid using certain materials.

Next Steps: Order all components and materials to test further.

WEEK 2-
Planned Tasks: Test the components and start creating the actual flow of the project physically.

Completed Tasks: Tested the MAX sensor, neopixel, and DC motor. Cannot test materials yet because they did not arrive.

Challenges Faced: The PAM amp was very difficult to test because all the wiring connected to it was temporary as we were still testing. Due to the floating wires, it was barely working. We also began testing the neopixel because it was a component we already had with us. It took us an insane amount of time to get the neoPixel to work because the one that we had was not working, so we thought it was an issue with the neopixel, and we borrowed around 4 more neopixels. We also tested the breadboard, in case that was the issue, and we tested around 3 power modules as well, in case ours was not working. Every test confused us more and more, because the neopixel would sometimes only turn on one led, despite the code being for all 16. Sometimes the neopixel turned on without the code running at all.

Solutions: At the end we used a borrowed neopixel and by power supply module #3, and the neopixel finally worked. The PAM amp stabilized enough for us to test it and determine that it was functioning. We did this by firmly taping all the connections for it and trying not to move it at all while we tested it. We could not get it soldered sooner because the circuit layout was still being developed.

WEEK 3-

Planned Tasks: Final system integration, 3D printing the enclosure, and stabilizing the power supply.

Completed Tasks: Assembled the final hardware inside the Blender-designed box and synchronized the pentatonic audio scale with the LED pulse.

Challenges Faced: There were numerous technical failures as the electromagnet and motor ideas failed completely at the last minute. The high power draw from the output components caused the combined setup to fail numerous times as the power supply was not enough for it. We also did not receive the materials for the ferrofluid until the very last couple of days so we found an alternative for the sound visualizer element, which was Chladni patterns. We used a metal plate and placed it on top of the speaker. The plate had a light layer of sugar grains on it. The concept was that the sound from the speaker, at a certain frequency, would cause the grains to vibrate and move into patterns known as Chladni patterns. This worked but it did not form distinct patterns. It was still our backup for the ferrofluid demo in case the materials did not arrive.

Solutions: We focused all our remaining energy on getting the Pulse Sensor —> NeoPixel —> Sound Module loop to work. We used a Buck Converter to isolate the power and spent hours debugging the I2C data stream to ensure the demo was stable.